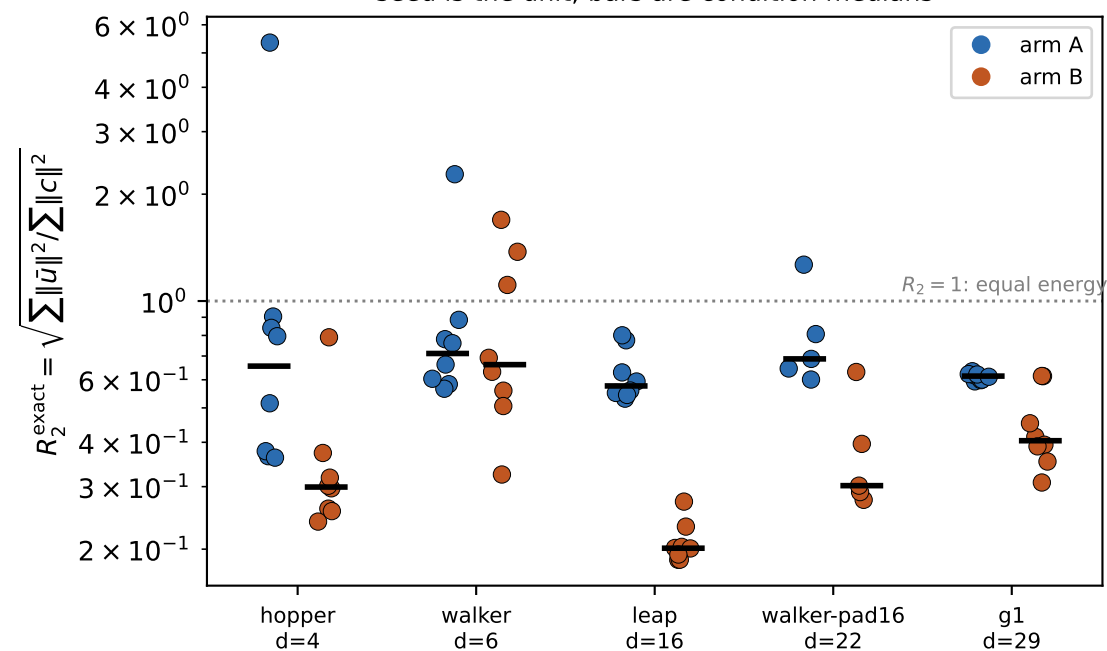
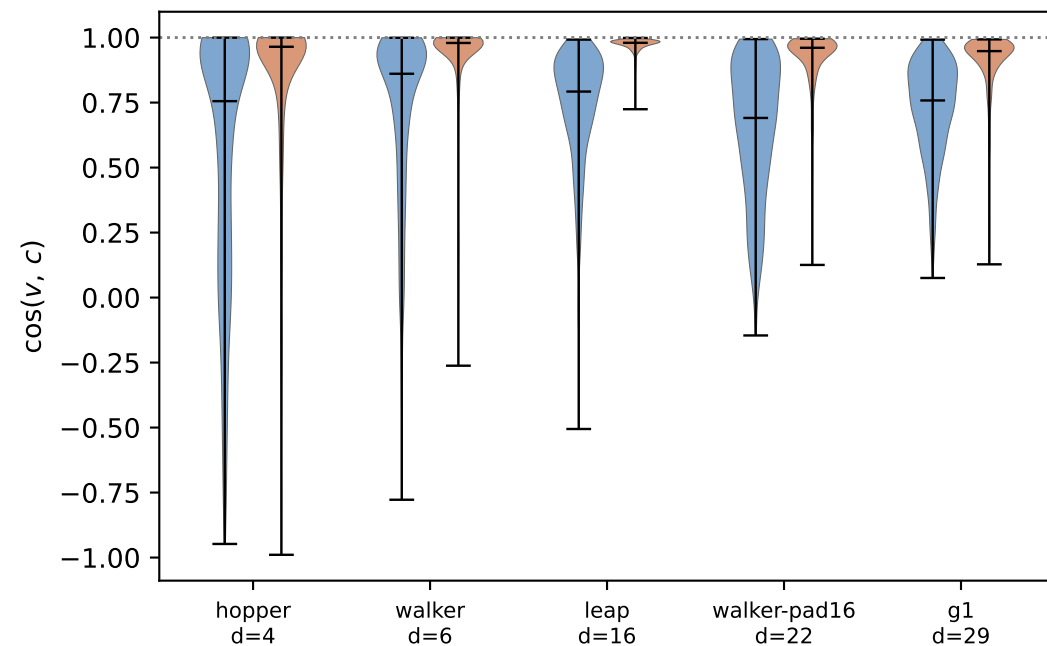


The uniform empirical-mean term in the implemented E-step (frozen checkpoints, read-only)

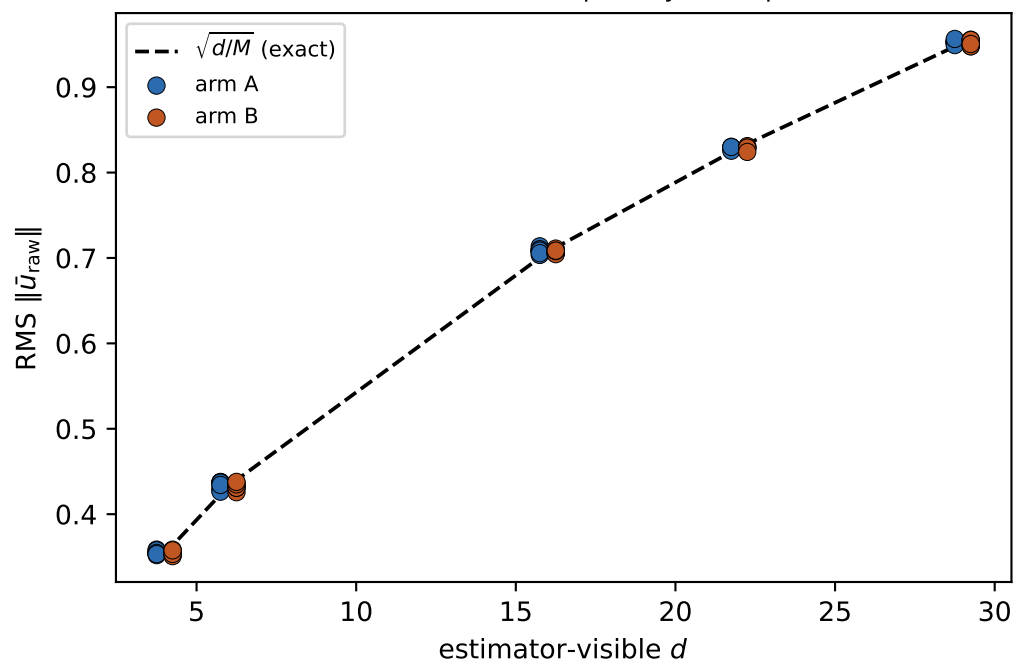
(a) Uniform vs centered energy, per checkpoint
seed is the unit; bars are condition medians



(b) Does $\tilde{u}$ move the direction? state-level $\cos(v, c)$
 $\cos = 1$ would mean $\tilde{u}$ changes nothing



(c) Raw-Gaussian probe sanity check
RMS (not median) is the quantity $\sqrt{d/M}$ predicts



(d) First-order expansion degrades as the logits spread
points are states; both arms

